

CS 350 Module Four Lab Guide

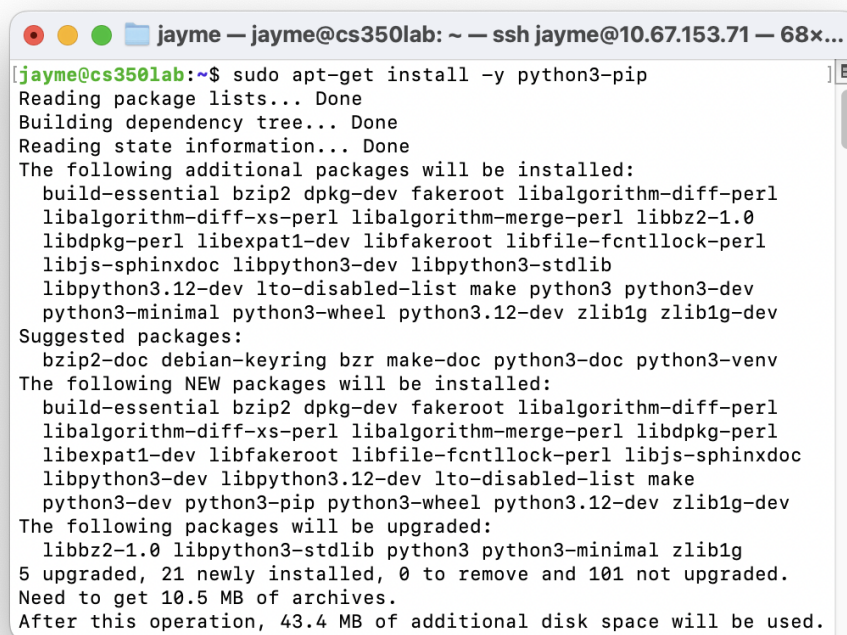
You have been working closely with the LED you added to your circuit in Module One. This was a good beginning, but design language often needs to present more information than what a simple LED can offer. This activity will show you how to expand your capabilities and offer this functionality to future programs.

In this lab, you will add another component to your solderless breadboard to provide an additional capability. The component you will be adding is a 16x2 display. 16x2 refers to the display's capability to provide 16 characters on each of two lines.

Step 1: Installing Packages

You will install some additional Python packages to work with the new hardware. The first package that you will install is the Python package manager.

```
sudo apt-get install -y python3-pip
```



```
jayme — jayme@cs350lab: ~ — ssh jayme@10.67.153.71 — 68x...
[jayme@cs350lab:~]$ sudo apt-get install -y python3-pip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  build-essential bzip2 dpkg-dev fakeroot libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libbz2-1.0
  libdpkg-perl libexpat1-dev libfakeroot libfile-fcntllock-perl
  libjs-sphinxdoc libpython3-dev libpython3-stdlib
  libpython3.12-dev lto-disabled-list make python3 python3-dev
  python3-minimal python3-wheel python3.12-dev zlib1g zlib1g-dev
Suggested packages:
  bzip2-doc debian-keyring bzip2-doc python3-doc python3-venv
The following NEW packages will be installed:
  build-essential bzip2 dpkg-dev fakeroot libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libdpkg-perl
  libexpat1-dev libfakeroot libfile-fcntllock-perl libjs-sphinxdoc
  libpython3-dev libpython3.12-dev lto-disabled-list make
  python3 python3-pip python3-wheel python3.12-dev zlib1g-dev
The following packages will be upgraded:
  libbz2-1.0 libpython3-stdlib python3 python3-minimal zlib1g
5 upgraded, 21 newly installed, 0 to remove and 101 not upgraded.
Need to get 10.5 MB of archives.
After this operation, 43.4 MB of additional disk space will be used.
```

Once you have pip3 installed, you need to set a variable that will allow you to install two Python modules needed to work with your display: `adafruit-blinka` and `adafruit-circuitpython-charlcd`.

Running the following script will set the variable to `true`, allowing you to use the pip3 tool to install these packages and make them available on your system.

```
sudo python3 -m pip config set global.break-system-packages true
```

```
jayme — jayme@cs350lab: ~ — ssh jayme@10.67.153.71 — 68x...
[jayme@cs350lab:~]$ sudo python3 -m pip config set global.break-system
-packages true
Writing to /root/.config/pip/pip.conf
jayme@cs350lab:~$
```

With this global configuration setting in place, you can now install each of your Python packages using the following command:

```
sudo pip3 install adafruit-blinka
```

```
jayme — jayme@cs350lab: ~ — ssh jayme@10.67.153.71 — 88x22
[jayme@cs350lab:~]$ sudo pip3 install adafruit-blinka
[sudo] password for jayme:
Collecting adafruit-blinka
  Downloading Adafruit_Blinka-8.47.0-py3-none-any.whl.metadata (5.5 kB)
Collecting Adafruit_PlatformDetect>=3.70.1 (from adafruit-blinka)
  Downloading Adafruit_PlatformDetect-3.74.0-py3-none-any.whl.metadata (3.8 kB)
Collecting Adafruit_PureIO>=1.1.7 (from adafruit-blinka)
  Downloading Adafruit_PureIO-1.1.11-py3-none-any.whl.metadata (3.0 kB)
Collecting binho-host-adapter>=0.1.6 (from adafruit-blinka)
  Downloading binho_host_adapter-0.1.6-py3-none-any.whl.metadata (2.0 kB)
Collecting pyftdi>=0.40.0 (from adafruit-blinka)
  Downloading pyftdi-0.55.4-py3-none-any.whl.metadata (3.2 kB)
Collecting adafruit-circuitpython-typing (from adafruit-blinka)
  Downloading adafruit_circuitpython_typing-1.11.0-py3-none-any.whl.metadata (3.2 kB)
Collecting sysv-ipc>=1.1.0 (from adafruit-blinka)
  Downloading sysv_ipc-1.1.0.tar.gz (99 kB)
    99.4/99.4 kB 497.3 kB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: pyserial in /usr/lib/python3/dist-packages (from binho-host-adapter>=0.1.6->adafruit-blinka) (3.5)
Collecting pyusb!=1.2.0,>=1.0.0 (from pyftdi>=0.40.0->adafruit-blinka)
  Downloading pyusb-1.2.1-py3-none-any.whl.metadata (2.2 kB)
```

```
sudo pip3 install adafruit-circuitpython-charlcd
```



```
jayme — jayme@cs350lab: ~ — ssh jayme@10.67.153.71 — 88x22
[jayme@cs350lab:~]$ sudo pip3 install adafruit-circuitpython-charlcd
Collecting adafruit-circuitpython-charlcd
  Downloading adafruit_circuitpython_charlcd-3.4.10-py3-none-any.whl.metadata (5.6 kB)
Requirement already satisfied: Adafruit-Blinka in /usr/local/lib/python3.12/dist-packages
  (from adafruit-circuitpython-charlcd) (8.47.0)
Collecting adafruit-circuitpython-mcp230xx (from adafruit-circuitpython-charlcd)
  Downloading adafruit_circuitpython_mcp230xx-2.5.13-py3-none-any.whl.metadata (4.0 kB)
Requirement already satisfied: adafruit-circuitpython-busdevice in /usr/local/lib/python
  3.12/dist-packages (from adafruit-circuitpython-charlcd) (5.2.9)
Collecting adafruit-circuitpython-74hc595 (from adafruit-circuitpython-charlcd)
  Downloading adafruit_circuitpython_74hc595-1.4.4-py3-none-any.whl.metadata (3.8 kB)
Requirement already satisfied: adafruit-circuitpython-typing~=1.5 in /usr/local/lib/pyth
  on3.12/dist-packages (from adafruit-circuitpython-charlcd) (1.11.0)
Requirement already satisfied: adafruit-circuitpython-requests in /usr/local/lib/python3
  .12/dist-packages (from adafruit-circuitpython-typing~=1.5->adafruit-circuitpython-charl
  cd) (4.1.6)
Requirement already satisfied: typing-extensions~=4.0 in /usr/local/lib/python3.12/dist-
  packages (from adafruit-circuitpython-typing~=1.5->adafruit-circuitpython-charlcd) (4.12
  .2)
Requirement already satisfied: Adafruit-PlatformDetect>=3.70.1 in /usr/local/lib/python3
  .12/dist-packages (from Adafruit-Blinka->adafruit-circuitpython-charlcd) (3.74.0)
Requirement already satisfied: Adafruit-PureIO>=1.1.7 in /usr/local/lib/python3.12/dist-
```

Step 2: Power Off Your Raspberry Pi

Your Raspberry Pi must be shut down and unplugged to continue. To safely shut down the operating system running on your Raspberry Pi, enter the following command from a terminal (or the console) of the Raspberry Pi:

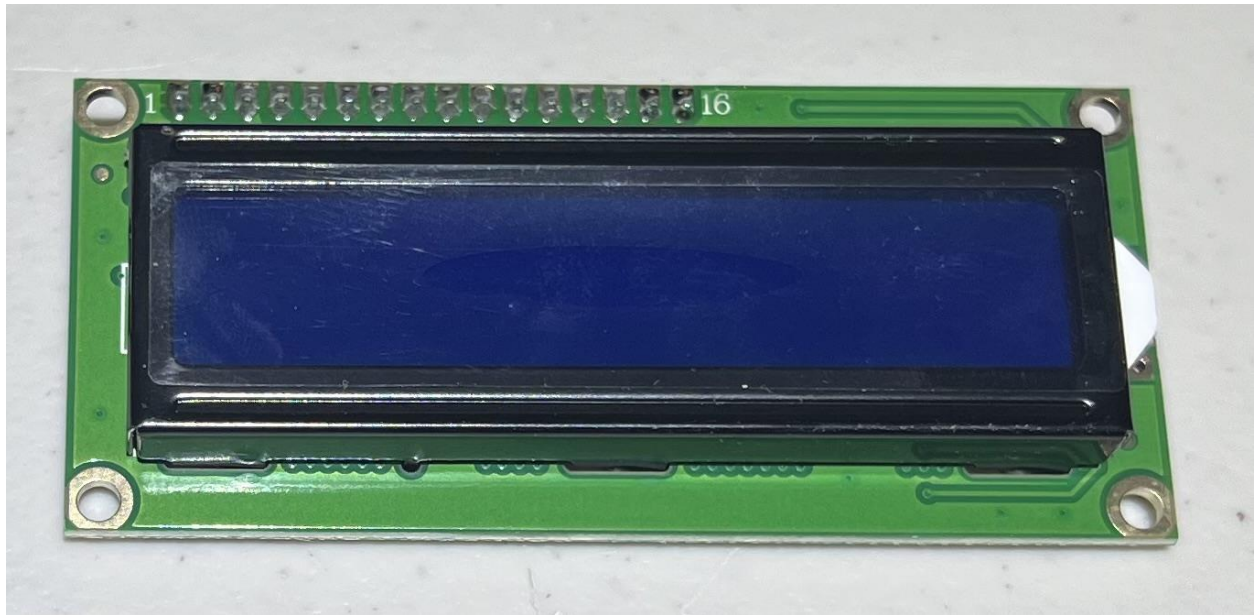
```
sudo halt
```

You will have to enter your password as you are executing the halt command with root privileges by calling it with *sudo*. Wait at least two minutes after your terminal window closes before unplugging your Raspberry Pi.

Step 3: Building the Circuit

Components

- 16x2 display
- Potentiometer
- 14 x male-to-male jumper cables
- The existing circuit you have been working with since Module One

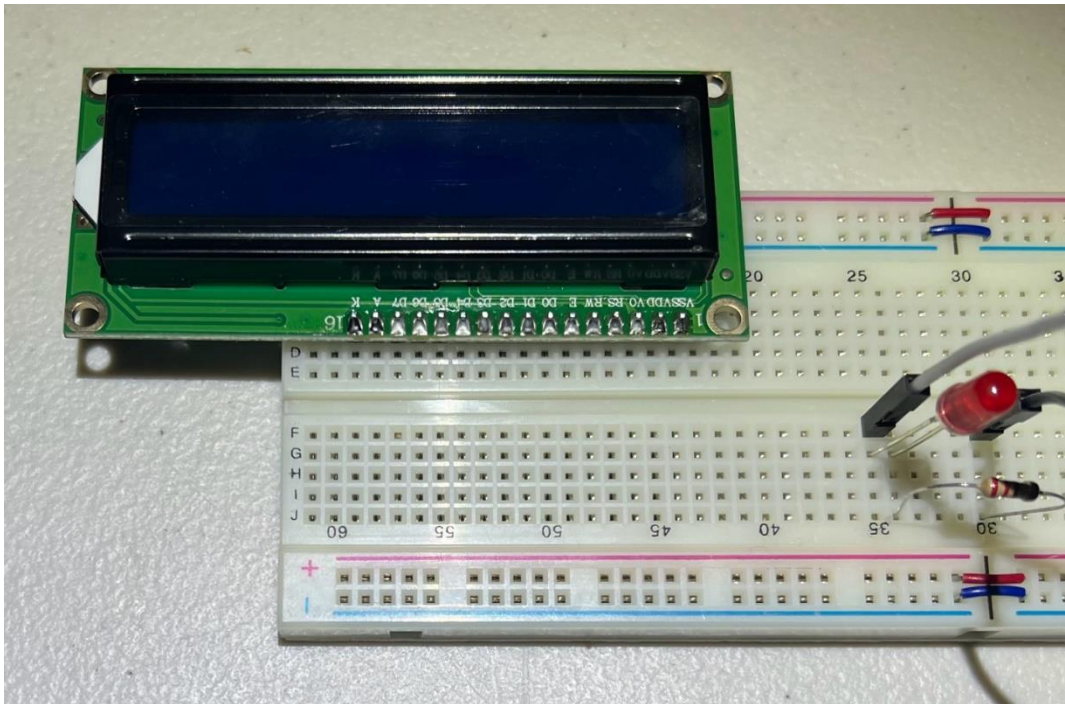


16 x 2 display

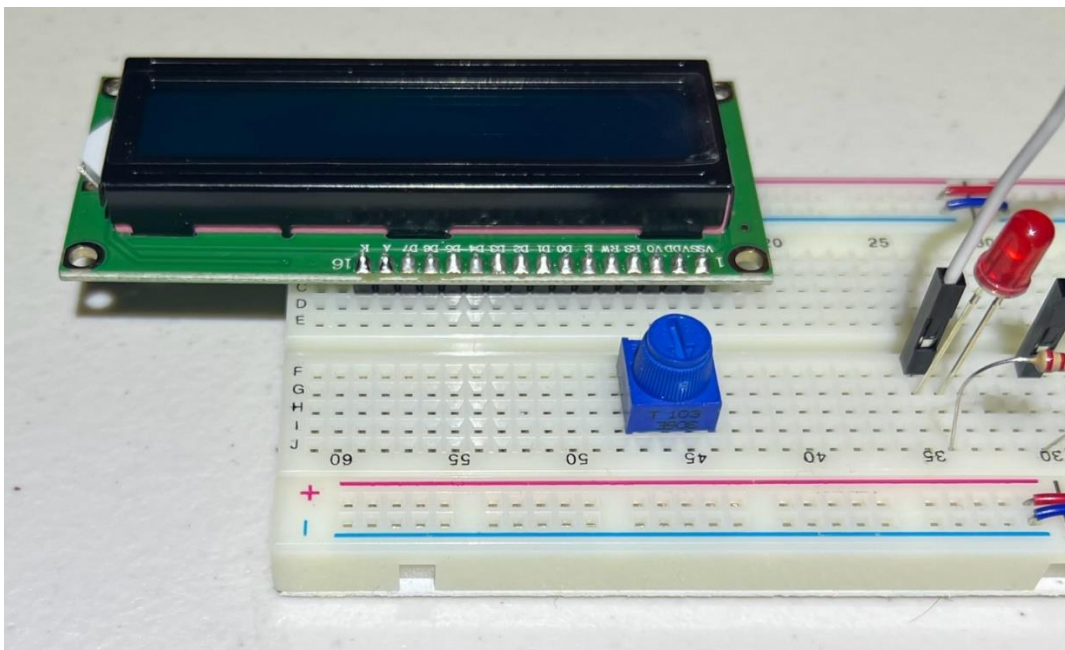


Potentiometer

Begin by placing the display on the solderless breadboard. To make room for the remaining elements of your circuit throughout this course, you will place the display on the board so that Pin 1 is in **C-44** in the orientation seen in the photo, with the rest of the pins along the same column.

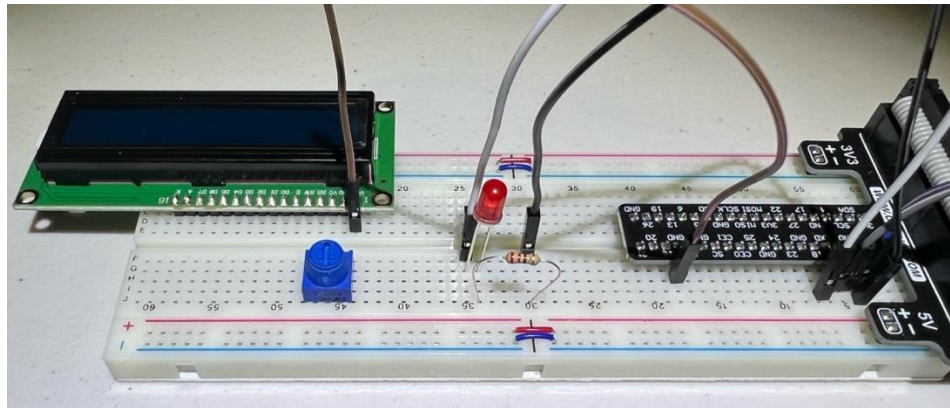


Next, place the potentiometer in Column G from Rows 45–47.

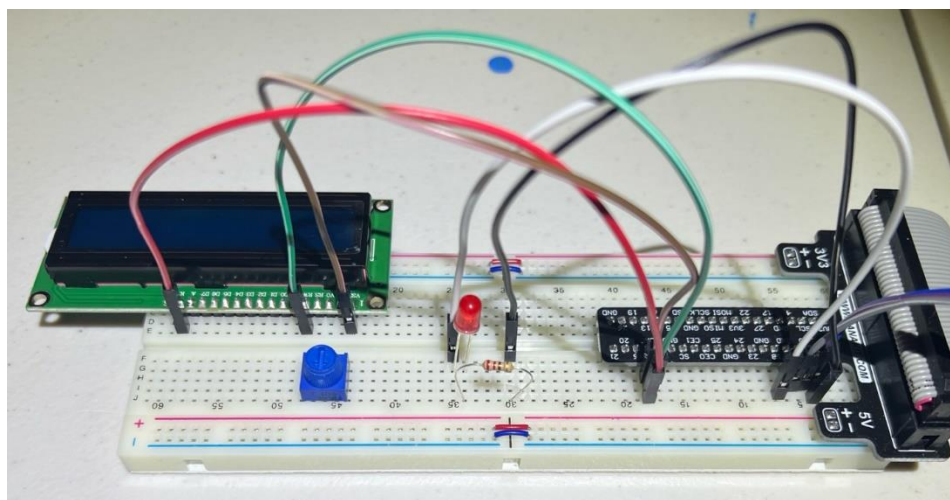


Several jumper wires must be connected for this circuit. You should begin by connecting the four wires connecting to ground.

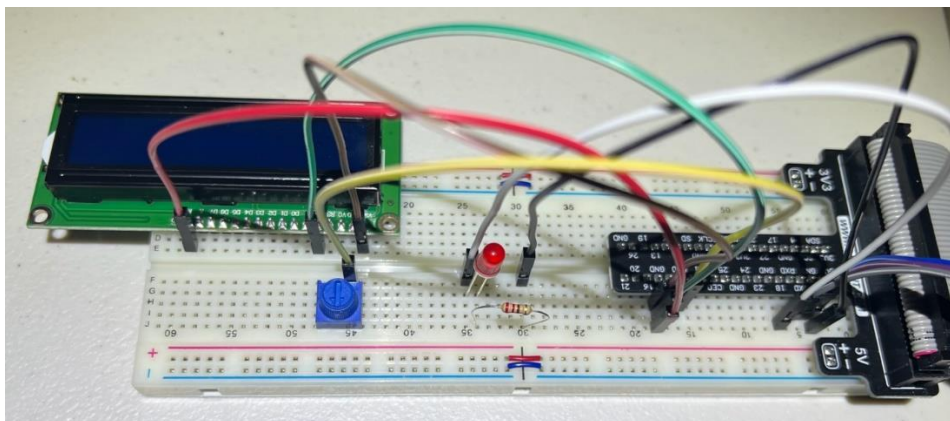
Connect a jumper wire from Pin 1 of the LCD (Column E, Row 44) to Column H, Row 18. The following image shows the correct installation:



Connect two more jumper wires: one from Pin 5 of the LCD (Column E, Row 48) to Column I, Row 18, and the second from Pin 16 of the LCD (Column E, Row 59) to Column J, Row 18. The following image shows the correct installation:



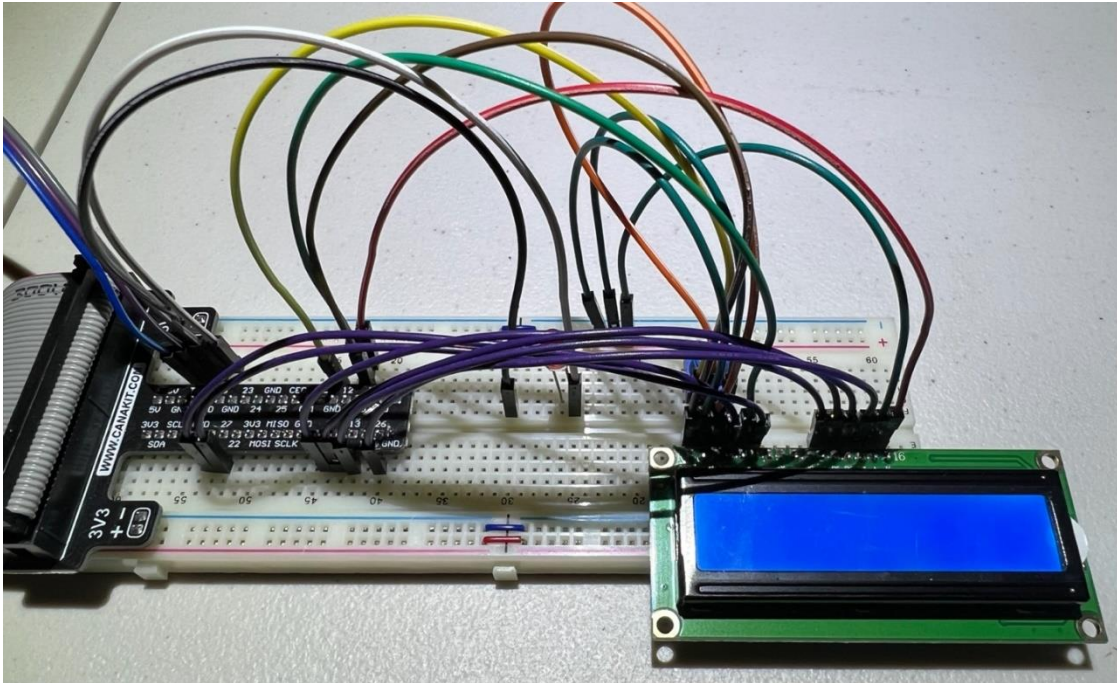
Connect a jumper wire from Pin 1 of the potentiometer (Column F, Row 45) to Column H, Row 16.



- Connect a jumper wire from Pin 4 of the LCD (RS) (Column E, Row 47) to Column C, Row 53 (GPIO 17).
- Connect a jumper wire from Pin 6 of the LCD (EN) (Column E, Row 49) to Column C, Row 52 (GPIO 27).
- Connect a jumper wire from Pin 11 of the LCD (D4) (Column E, Row 54) to Column C, Row 44 (GPIO 5).
- Connect a jumper wire from Pin 12 of the LCD (D5) (Column E, Row 55) to Column C, Row 43 (GPIO 6).

- Connect a jumper wire from Pin 13 of the LCD (D6) (Column E, Row 56) to Column C, Row 42 (GPIO 13).
- Connect a jumper wire from Pin 14 of the LCD (D7) (Column E, Row 57) to Column C, Row 40 (GPIO 26).

Note: It's important to keep track of which connections go to which GPIO pins, as they will be used to control what is displayed on the LCD panel. Knowing which connections go to which GPIO pins will dictate how you configure the software to write to the display.

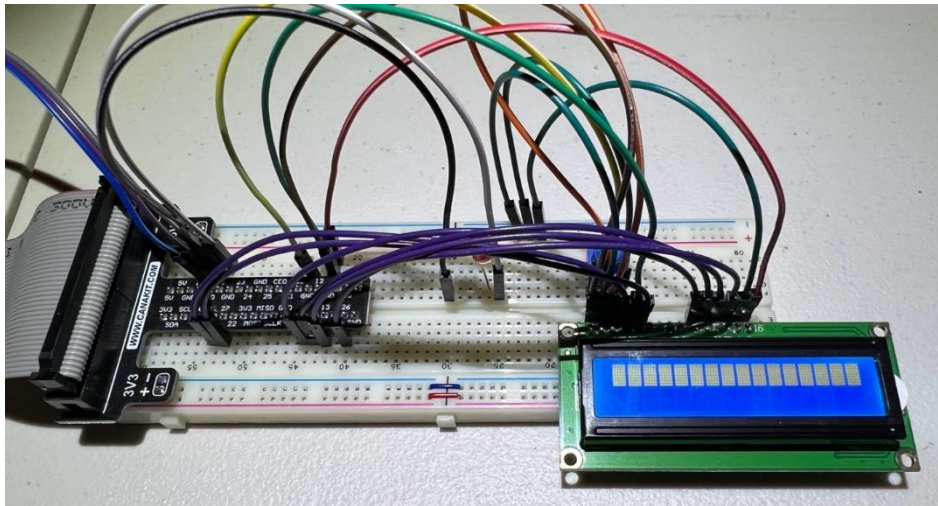


Your circuit should now be complete. Reach out to your instructor if you need help completing the circuit.

Step 4: Power Up Your Raspberry Pi

Reconnect your Raspberry Pi to power, and once it has booted, log in.

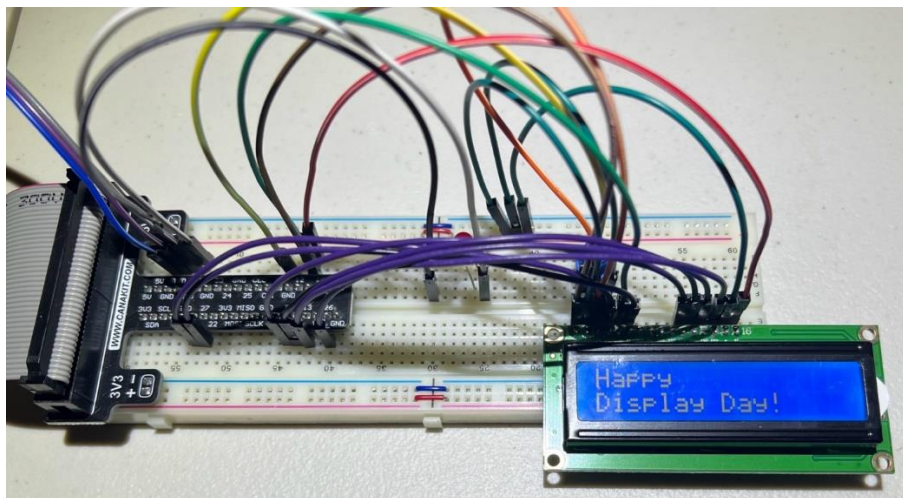
At this point, you can adjust the potentiometer by carefully rotating it to change the contrast on the LCD display. If you see blocks on the screen, like in the image below, use the potentiometer to adjust the contrast so that the display is all one color.



Step 5: Testing Your Display

You can test your display using the DisplayTest.py script that is part of the source code for Module Four. You can run this test program with the following command:

```
sudo python3 DisplayTest.py
```



You can now examine the code in the DisplayTest.py file. Feel free to modify the loop to try to display different information. Remember, you have 16 characters on each line, so make them count.

Lab Questions

Answer these questions in a separate Word document and include it as part of your submission for the Module Four assignment.

1. Why do you have a sleep command in your loop?
2. What is the purpose of having a text display on an embedded device?
3. How can you think of the display device as something that could relate to a state machine?